

TrES-3b

R-2537

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110427 · created 2026-08-25T23:48:39+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2455678.8835 +/- 0.00089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.011
Transit depth (Rp/Rs)^2	1.78 +/- 0.29 [%]
Orbital Inclination (inc)	86.92 +/- 0.74
Airmass coefficient 1 (a1)	1.021 +/- 0.01
Airmass coefficient 2 (a2)	-0.017 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.02 %
Best Comparison Star	#2 - [462, 392]
Optimal Aperture	4.16
Optimal Annulus	8.65
Transit Duration (day)	0.0786 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.133 $\pm$ 0.011 vs 0.16309 (deviation 2.74 σ)
Duration fit vs published	0.0786 d vs 0.0567 d (deviation 11.54 σ)
Mid-time O-C	1.4 min
Depth SNR	12.1
Residual scatter	1.02 %

Light curve

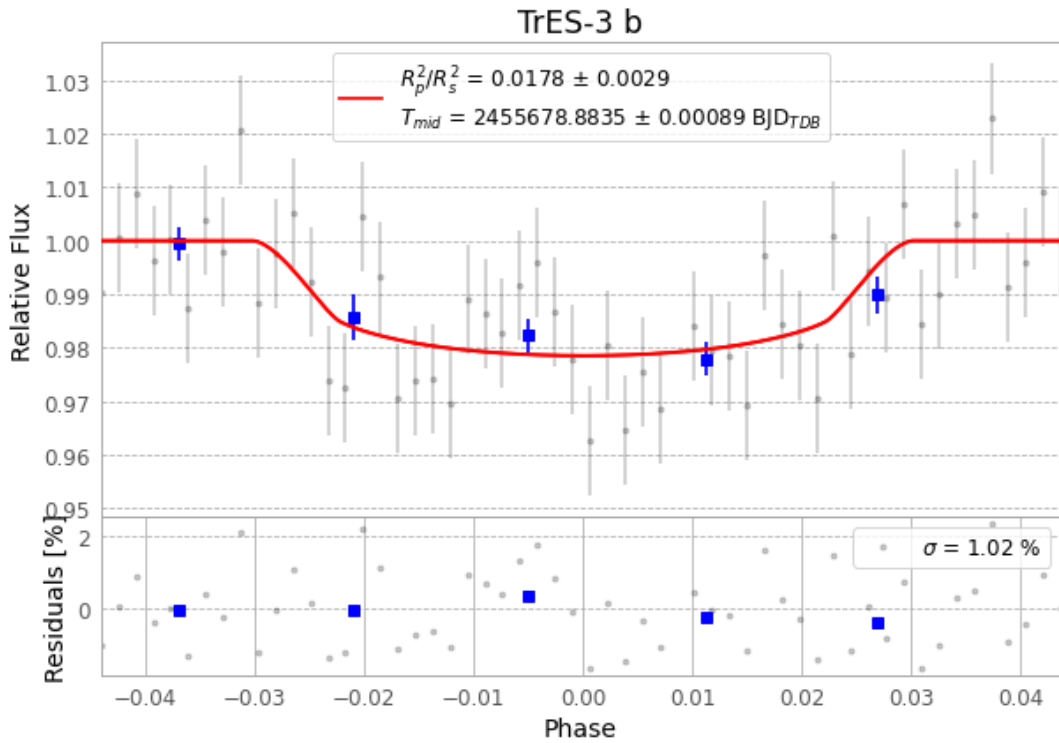


Figure 1 — FinalLightCurve_TrES-3 b_2011-04-27.png (SHA-256 eb3c280c53a53eba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 171], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 686085bb0372d67c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), TRES-3110427074512.FITS → TRES-3110427103013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110427.log (bf65b70b872c...), FinalLightCurve_TrES-3 b_2011-04-27.png (eb3c280c53a5...), FinalLightCurve_TrES-3 b_2011-04-27.csv (87955455a502...)

Compute resources

Wall time 302.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:48Z.